

# Assignment 1

a.

Aim. Write a program in C for matrix subtraction.  
Show the statement level calculation of time complexity  
in the right column. Write the expression for  
order of time complexity

Time complexity

```
#include <stdio.h>
```

```
void main() {
```

```
    int a[3][3];
```

1

```
    int b[3][3];
```

1

```
    printf("A array");
```

$C_1$

```
    for(int i=0; i<3; i++){
```

$1 + 3 \times 2$

```
        for(int j=0; j<3; j++){
```

$3 \times (1 + 3 \times 2)$

```
            scanf("%d", &a[i][j]);
```

$(C_2 + 3) \times 3 \times 3$

```
        }
```

```
    }
```

```
    printf("B array");
```

$C_1$

```
    for(int i=0; i<3; i++){
```

$1 + 3 \times 2$

```
        for(int j=0; j<3; j++){
```

$3 \times (1 + 3 \times 2)$

```
            scanf("%d", &b[i][j]);
```

$(C_2 + 3) \times 3 \times 3$

```
        }
```

```
    }
```

```
    for(int i=0; i<3; i++){
```

$1 + 3 \times 2$

```
        for(int j=0; j<3; j++){
```

$3 \times (1 + 3 \times 2)$

```
            a[i][j] = a[i][j] - b[i][j];
```

$(9 + 2) \times 9$

```
        }
```

```
    }
```

```

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        printf("%d", a[i][j]);
    }
    printf("\n");
}

```

$1 + 2 \times 3$   
 $3(1 + 2 \times 3)$   
 $(c_1 + 3) \times 9$   
 $c_1 \times 3$

Time complexity for  $3 \times 3$  matrix  
 $= 12c_1 + 18c_2 + 296$

For  $n \times n$  matrix

Time complexity =  $1 + 1 + c_1 + 1 + n \times 2 + n(1 + 2n) +$   
 $(c_2 + 3)n^2 + c_1 + 1 + 2n + n(1 + 2n) + (c_2 + 3)n^2$   
 $+ 1 + 2n + n(1 + 2n) + n^2(11)$   
 $+ 1 + 2n + n(1 + 2n) + (c_1 + 3)n^2 + nc_1$

$$= \cancel{6} + \cancel{2c_1} + \cancel{nc_1} + \cancel{6n^2} + \cancel{c_1n^2} + 3n^2$$

The order of this expression is  $O(n^2)$

# Assignment 1

b.

Aim: Write a program in C to show the region of memory used by code, data, stack and extra segments. Calculate the space required for the program. Write the expression for order of space complexity.

```
#include <stdio.h>
void f(int a){
    printf("In Local variable of caller f has value = %d\n", a, &a);
    if(--a){
        f(a);
    }
}
void main(){
    int z = 4;
    printf("In Local variable z of caller function\n", z, &z);
    f(z);
}
```

| function | variable | Value | address          |
|----------|----------|-------|------------------|
| main     | a        | 4     | 000000000 61FE4C |
| f        | a        | 4     | 000000000 61FE20 |
| f        | a        | 3     | 000000000 61FDFO |
| f        | a        | 2     | ↓ 61FD00         |
| f        | a        | 1     | 61FD90           |

Hex byte difference = 30 , 48 bytes

extra segments used & data segment size is zero  
space used by code <sup>segment</sup> = ~~82 KB~~ 13.8 KB